Formalism for the density matrix in an ADM $3+1$ foliation of spacetime

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CONTENTS

I. INTRODUCTION

In Ref. [1] we discussed a thermal equivalence principle that should hold when dealing with a statistical physics theory [2–4] of a many body fluid within the framework of general relativity: Einstein field equations $G_{\alpha\beta} = 8\pi T_{\alpha\beta}$ suggest that the statistical properties of spacetime, i.e. the left hand side of Einstein equations, and the statistical properties of the matter moving in it, i.e. the right hand side of Einstein equations, should be equivalent. In particular when looking at the right hand side one reaches for the density matrix of the many (scalar) particle system

$$\hat{\rho} = \exp\left(-\int_{\partial\Omega^L} \hat{T}^{\alpha\beta}\,\beta_\alpha\,dS_\beta\right), \tag{1.1}$$

that is Eq. (5.1) in Ref. [1]. In this equation we use the hat to denote the corresponding operator, $\partial\Omega^L$ is an arbitrary spacelike hypersurface bounding the $4L$–volume Ω^L occupied by the L particles, and β_α is a 4–vector such that $\sqrt{\beta_\alpha\beta^\alpha} = 1/k_BT$ with k_B Boltzmann constant and T the absolute temperature.

Within Arnowitt, Deser, and Misner (ADM) $3+1$ foliation of spacetime [4] we can evaluate the integral at the exponent of Eq. (1.1) choosing an hypersurface orthogonal to the time component so that $dS_0 = \prod_{\mu=1}^L \sqrt{{}^3g(\boldsymbol{r}_\mu)}\,d\boldsymbol{r}_\mu$ where ${}^3g = \det\{g_{ij}\} > 0$ with g_{ij} the metric on the spatial components and $\boldsymbol{r}_\mu$ the *shifted* spatial positions of the μ particle, with $d\boldsymbol{r}_\mu = d\boldsymbol{x}_\mu + \boldsymbol{N}dt_\mu$. Here $\tilde{x}_\mu = (\boldsymbol{x}_\mu, t_\mu)$ is the 4–vector event for particle μ and $\boldsymbol{N}$ is the *shift* in the ADM foliation

$$||g_{\alpha\beta}|| = \begin{pmatrix} -(N^2 - N^iN_i) & N_i \\ N_i & g_{ij} \end{pmatrix}, \tag{1.2}$$

where N is the *lapse* and N^i the ith spatial component of the shift (see Eq. (4) in [4]). We will generally use a Greek index to denote one of the 4 spacetime components and a Roman index to denote one of the 3 spatial components. A bold face symbol denotes its status as a three dimensional vector.

By the definition of the stress-energy tensor

$$\int \hat{T}^{00}\,dS_0 = \hat{\mathscr{H}}_L, \tag{1.3}$$

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where $\hat{\mathscr{H}}_L$ is the Hamiltonian operator of the L particles. For a fluid invariant under translations and rotations this is the only stress-energy tensor component entering the density matrix in Eq. (1.1) [5].

Assuming $\beta_0 = 1/k_B\tilde{T}$ constant in the chosen constant time section of spacetime, we reach the following expression

$$\hat{\rho} = \exp\left(-\beta_0\hat{\mathscr{H}}_L\right). \tag{1.4}$$

The coordinates, $R = \{\boldsymbol{r}_1, \ldots, \boldsymbol{r}_L\}$, representation of the density matrix (1.4) can then be expanded into a path integral [6]

$$\rho(R, R'; \beta_0) = \int \rho(R, R_1; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} d^{3N}R_k, \tag{1.5}$$

where $\tau = \beta_0/M$ is a small imaginary *timestep*, and the *primitive approximation* [6] to the high temperature, $M \gg 1$, density matrix

$$\rho(R, R'; \tau) = \frac{\prod_{\mu=1}^N \sqrt{^3g(\boldsymbol{r}_\mu)}}{(4\pi\lambda\tau)^{3N/2}} \exp\left[-\sum_{\mu=1}^N g_{ij}(\boldsymbol{r}_\mu)(r_\mu^i - {r'}_\mu^i)(r_\mu^j - {r'}_\mu^j)/4\lambda\tau - \tau\mathscr{U}_N(R)\right], \tag{1.6}$$

where $\mathscr{H}_N = \mathscr{K}_N + \mathscr{U}_N$ with $\mathscr{U}_N$ the potential energy of the N particles and we assumed *non relativistic* [1] particles with a kinetic energy $\mathscr{K}_N = -\lambda\sum_{\mu=1}^N \nabla_\mu^2$ with $\lambda = \hbar^2/2m$. We also use Einstein convention to tacitly assume a sum over repeated indexes. In other words at each timeslice $t_k = k\tau$, for $k = 1, 2, \ldots, M-1$, we consider the section of spacetime at constant imaginary time and construct the representation of the high temperature density matrix on that spatial foil. Eq. (1.6) reduces to the usual expression in Euclidean space [6]. It has been found that one may need corrections due to the curvature of space and to the van Vleck determinant [7, 8]. Fully relativistic particles with a dispersion relation $\epsilon(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 + m^2}$ with $\boldsymbol{p} = -i\hbar\nabla$ are *non-local* since the Taylor expansion of ϵ in $\boldsymbol{p}$ has ever increasing powers of $\boldsymbol{p}$. In order to restore locality is therefore necessary to add for each particle its *antiparticle* [9] with negative energy $\bar{\epsilon}(\boldsymbol{p}) = -\sqrt{\boldsymbol{p}^2 + m^2}$.

In the high temperature regime one recovers classical statistical mechanics on a curved space. For $M = 1$ and $R \approx R'$

$$\rho(R, R'; \tau) \propto \prod_{\mu=1}^N \sqrt{^3g(\boldsymbol{r}_\mu)} \exp\left[-\tau\mathscr{U}_N(R)\right], \tag{1.7}$$

where $\tau = \beta_0$ is assumed to be small.

We have previously proposed [10] to consider the $\sqrt{^3g}$ factors as an external potential

$$U_N(R) = -\frac{1}{\tau}\ln\left(\prod_{\mu=1}^N \sqrt{^3g(\boldsymbol{r}_\mu)}\right) + \ldots, \tag{1.8}$$

where the dots stand for additional corrections due to the space curvature and the van Vleck determinant [7, 8].

The classical limit (1.7) has been used in various spatial dimensions to describe the properties of Coulomb systems. These offer some exact analytic solutions in one dimension [11, 12], in two dimensions [13], and at a special value of the coupling constant in two dimensions on various surfaces: The plane [14], the cylinder [15], the sphere [16], the pseudosphere [17], the Flamm's paraboloid [18, 19].

On the other hand, the path integral (1.5) with the primitive approximation (1.6) has been treated for example in Refs. [7, 8] and applied in various ways to various systems. For the example of a sphere see Refs. [20–22].

II. CONCLUSIONS

In order to extract the statistical mechanics properties of a many (scalar) particle fluid within general relativity, we propose to use a path integral where on each imaginary timeslice one selects a constant time foil of the $3+1$ ADM spacetime foliation.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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